



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

Artificial and Computational Intelligence

DSE ZG557

Contact Session 1

What is Artificial Intelligence?

Akhil P V

BITS Pilani WILP

Course scope: DSE ZG557 Digital Learning Handout, Part B, Contact Session 1.



Which of these systems is intelligent?

Which of these systems would you call AI?

Calculator

Thermostat

Spam filter

**Route-planning
application**

**Chess-playing
program**

**Conversational
assistant**

Think for 30 seconds

1. Which systems are intelligent?
2. What criterion did you use?
3. Would everyone give the same answer?

Reference: T1, Chapter 1, Exercise 1.7, p. 32.



The central question of the course

How can we build a system that can...



... and do these things effectively in a complex environment?

Example

An autonomous delivery system must observe traffic, represent routes, select a path, act safely and improve from experience.

Different parts of the course address different parts of this sequence.



Our journey across 16 contact sessions

The course develops an intelligent system step by step

- 1 AI foundations — CS1–CS2**
What is intelligence, and what is an intelligent agent?
- 2 Search and optimisation — CS3–CS7**
How can an agent find or improve a solution?
- 3 Game playing — CS8**
How should an agent act against an opponent?
- 4 Logical reasoning — CS9–CS10**
How can an agent represent facts and infer conclusions?
- 5 Uncertainty and time — CS11–CS13**
How can an agent reason when information is uncertain or changes over time?
- 6 Learning and responsibility — CS14–CS16**
How can an agent learn, explain and act responsibly?

Reference: DSE ZG557 Digital Learning Handout, Part B: Learning Plan.



Learning objectives for Contact Session 1

By the end of this session, you should be able to:

1. Explain why AI has more than one definition.
2. Distinguish:
 - **thinking humanly**
 - **acting humanly**
 - **thinking rationally**
 - **acting rationally**
3. Explain why this course emphasises rational action.
4. Identify the major disciplines that contributed to AI.
5. Recognise AI capabilities in real application domains.

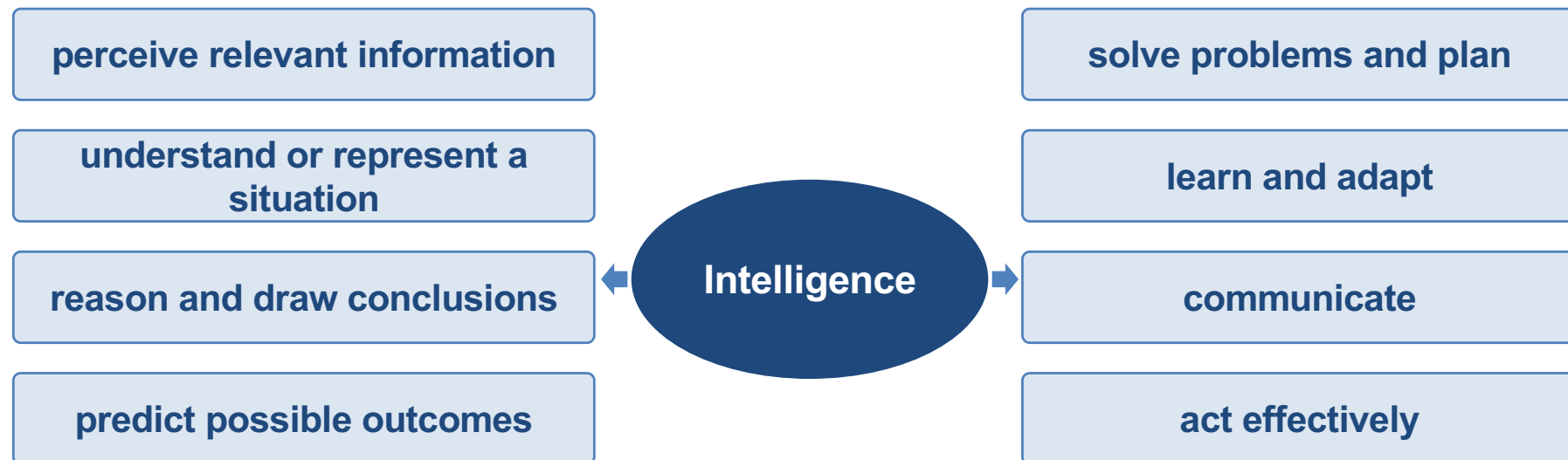
Today's path



Reference: DSE ZG557 Handout, CS1; T1, §§1.1, 1.2 and 1.4.

What do we mean by intelligence?

Intelligence involves the ability to:



Important

A system may demonstrate some of these abilities without demonstrating all of them.



What is Artificial Intelligence?

Artificial Intelligence

The scientific and engineering study of understanding and building intelligent entities.

AI includes systems that perform tasks involving:

learning and perception

reasoning and problem
solving

planning and decision-
making

language and communication

action in physical or digital
environments

AI is a broad field — not a single algorithm.

Reference: T1, Chapter 1, p. 1.

Two questions produce four approaches

Question 1

What do we study?

Internal thought or reasoning

OR

External behaviour or action?

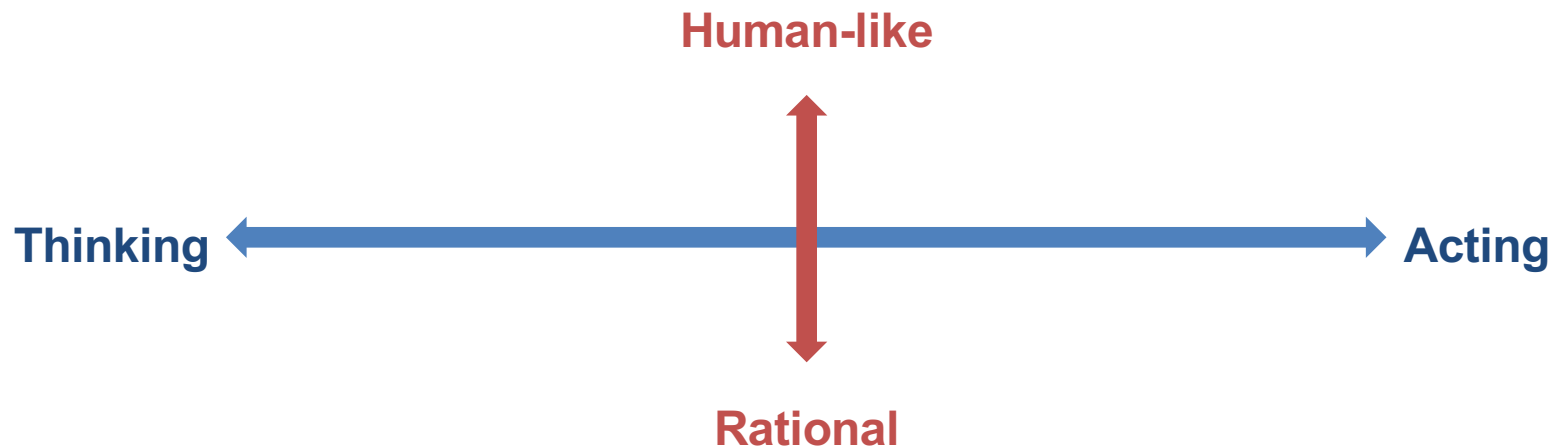
Question 2

What is the standard of success?

Human performance

OR

Rational performance?



Combining these questions gives four approaches to AI.

Four major approaches to AI

	Human standard	Rational standard
Thinking	COGNITION Thinking humanly Model how humans actually think	LOGIC Thinking rationally Use correct reasoning principles
Acting	DIALOGUE Acting humanly Behave like an intelligent human	OBJECTIVE Acting rationally Choose actions that produce the best expected outcome

The approaches ask different questions and require different evidence.

Reference: T1, Figure 1.1 and §1.1, pp. 1–2.



Thinking humanly: cognitive modelling

Goal	Methods	Evaluation
Model the processes by which humans actually think.	How can human thinking be studied? • Introspection • Psychological experiments • Brain imaging	Does the program's reasoning process resemble the process followed by human subjects?

Illustrative example

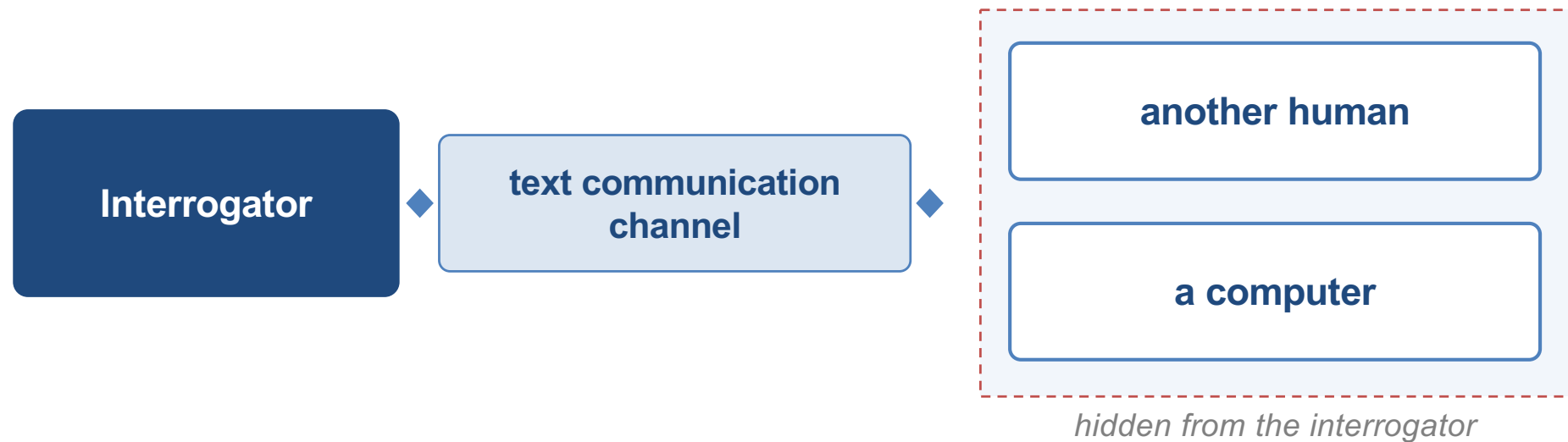
Newell and Simon's General Problem Solver compared computer reasoning traces with human problem-solving traces.

Reference: T1, §1.1.2, p. 3.

Acting humanly: the Turing Test

Turing Test

A human interrogator communicates through text with another human and a computer, without knowing which is which.



The computer succeeds when the interrogator cannot reliably distinguish its responses from those of the human.

Focus: observable human-like behaviour

Reference: T1, §1.1.1, pp. 2–3.



Capabilities needed for the Turing Test

To act humanly in the Turing Test, a system needs:

1. Natural Language Processing — communicate in human language

2. Knowledge Representation — store information about the world

3. Automated Reasoning — answer questions and derive conclusions

4. Machine Learning — adapt to new situations and patterns

Total Turing Test also requires

5. Computer vision

6. Robotics

Reference: T1, §1.1.1, pp. 2–3.

Strengths and limitations of acting humanly

Strengths

- Provides an observable operational test
- Integrates several AI capabilities
- Avoids requiring a philosophical definition of thought

Limitations

- Imitating humans is not the same as understanding intelligence
- Human behaviour is not always correct or rational
- Passing a test does not explain how the system works
- It does not directly provide a method for designing intelligent systems

Key question

Should an aircraft imitate a bird — or apply the principles of aerodynamics?



Thinking rationally: the laws of thought

Goal

Use formal rules to derive correct conclusions from available premises.

Historical foundation

Aristotle's syllogisms provided patterns for valid arguments.

Primary tool
LOGIC

Example structure

Premise 1

+

Premise 2



Conclusion

Reference: T1, §1.1.3, p. 4.



Rational reasoning: example and obstacles

Example

1. All safety-critical alerts require immediate review.
2. Alert A is safety-critical.
- 3. Therefore, Alert A requires immediate review.**

Two major obstacles

1

Knowledge representation

How do we convert informal, incomplete or uncertain knowledge into precise logical statements?

2

Computational feasibility

A logically solvable problem may still require too much time or memory.

In principle $\neq$ In practice



Acting rationally: the rational-agent approach

A rational agent chooses an action that is expected to produce

THE BEST OUTCOME

or, under uncertainty, **THE BEST EXPECTED OUTCOME**

Information

Possible actions

Expected
outcomes

Selected action

The decision depends on

available information

possible actions

expected
consequences

the performance
objective

Focus

Doing the appropriate thing — not necessarily behaving like a human.

Reference: T1, §1.1.4, pp. 4–5.



Why does the textbook prefer rational action?

The rational-agent approach is:

More general

Correct logical inference is one possible way to act rationally — but not the only way.

Action-oriented

A system may need to act even when no perfectly correct conclusion is available.

Measurable

Performance can be evaluated using an explicit objective.

Scientifically useful

Rationality can be analysed using mathematics and engineering.

Reality

Perfect rationality may be impossible under limited time and computation.

Reference: T1, §1.1.4, pp. 4–5.



Comparing the four approaches

Approach	Central question	Success means
Thinking humanly	Does the internal process resemble human thought?	Match evidence about human cognition
Acting humanly	Does the behaviour appear human?	Indistinguishable or convincingly human-like behaviour
Thinking rationally	Are the conclusions logically correct?	Valid reasoning from given premises
Acting rationally	Does the action achieve the best expected result?	High performance according to an objective

Do not confuse the process with the outcome.

Reference: T1, §1.1, pp. 1–5.

Activity: classify the approach

Match each system description to the most relevant approach

A

A model reproduces the characteristic mistakes humans make in mental arithmetic.

B

A conversational system is difficult to distinguish from a person through text.

C

A theorem prover derives a conclusion using valid inference rules.

D

A navigation system chooses the route with the lowest expected travel time.

Choices

Thinking humanly

Acting humanly

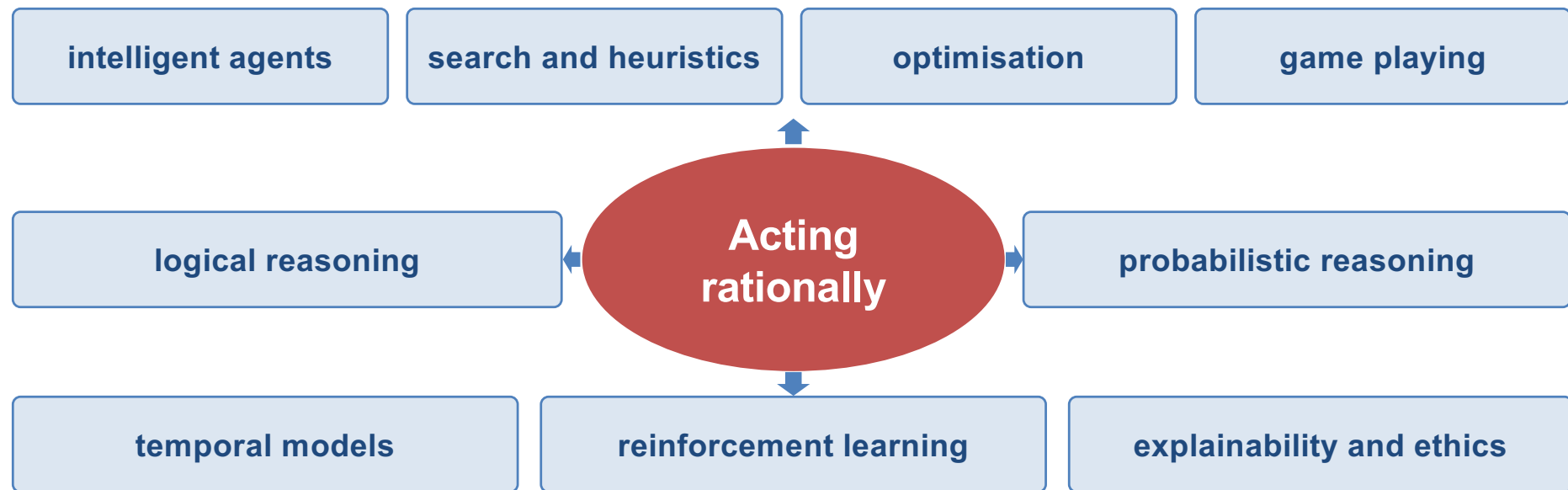
Thinking rationally

Acting rationally

Reference: T1, §1.1, pp. 1–5.

The perspective used in this course

This course mainly studies systems that act rationally. To build rational behaviour we will study:



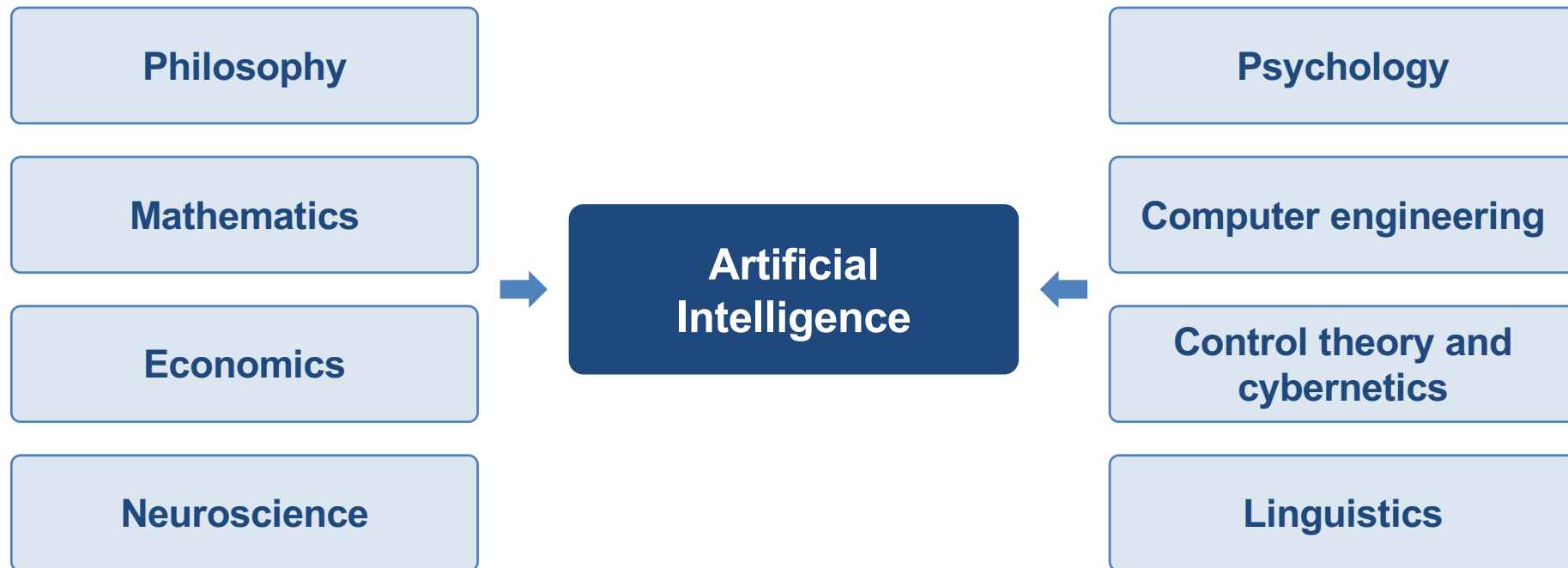
Next session

Agents, environments and rationality

Reference: T1, §1.1.4, pp. 4–5; DSE ZG557 Handout, Part B.

AI did not emerge from one discipline

Disciplines contributing to AI



Each discipline contributed a different question, formalism or engineering capability.

Philosophy and mathematics

Philosophy asks:

- Can formal rules produce valid conclusions?
- How does knowledge arise?
- How does knowledge lead to action?
- Is the mind a physical computational system?

Mathematics contributes:

- formal logic
- algorithms and computability
- complexity and tractability
- probability for uncertain reasoning

Connection to this course
Logic + Search + Probability



Economics and rational decision-making

Economics asks:

- How should an agent choose among preferred outcomes?
- How should it decide under uncertainty?
- What happens when other agents affect the outcome?
- How should it value rewards received in the future?

Major contributions:

- Utility theory
- Decision theory
- Game theory
- Sequential decision-making
- Markov Decision Processes

Decision theory = Probability + Utility

Reference: T1, §1.2.3, pp. 9–10.

Neuroscience and psychology

Neuroscience

How does the brain process information?

Contributions:

- neurons and neural communication
- biological learning mechanisms
- perception and brain organisation

Psychology

How do humans and animals think and act?

Contributions:

- experimental study of behaviour
- cognitive models
- memory, perception and problem solving
- the brain as an information-processing system

Important distinction

AI models may be inspired by humans without exactly reproducing the human brain.

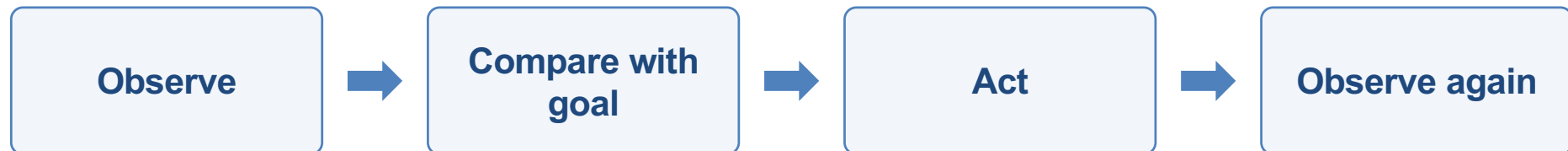
Computer engineering and control theory

Computer engineering contributes

- computational hardware
- memory and processing
- programming languages
- software systems
- sensors and actuators

Control theory and cybernetics contribute

- feedback
- goal-directed regulation
- minimising the difference between current and desired states
- maximising an objective over time



feedback loop – the cycle repeats continuously

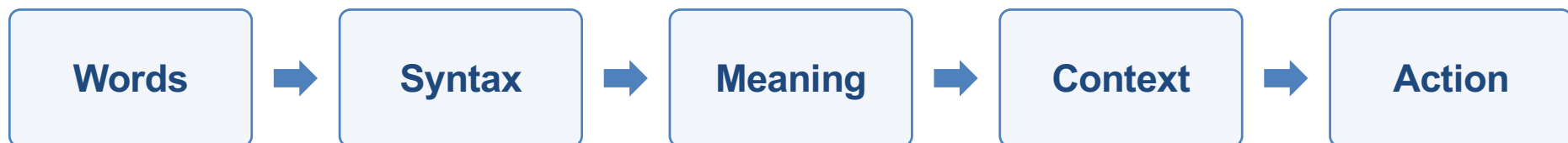
Linguistics and language intelligence

Linguistics asks
How does language relate to thought and knowledge?

Contributions to AI

- formal models of language structure
- syntax and grammar
- computational linguistics
- natural language processing
- recognition that meaning depends on context and world knowledge

Language understanding is more than matching words.



How the foundations reappear in this course

Foundation	Course connection	Sessions
Philosophy	Rationality, knowledge and action	CS1–CS2
Mathematics	Search, logic, probability and complexity	CS3–CS7
Economics	Games, utility, MDPs and decisions	CS8
Neuroscience and psychology	Learning, cognition and perception	CS11–CS15
Computer engineering	Implementation of intelligent systems	Throughout
Control theory	Feedback, objectives and sequential action	CS13–CS15
Linguistics	Language representation and communication	CS9–CS10

The foundations are not separate history – they support the algorithms we will study.

Reference: T1, §1.2, pp. 5–16; DSE ZG557 Handout, modular content structure.

From components to an intelligent system

An intelligent application may combine:



Learning from data or experience influences multiple stages

Modern AI systems often integrate several components rather than relying on one method.

Reference: T1, Chapter 1 overview and §1.4, pp. 1 and 28–29.

Application categories in the textbook

T1 Section 1.4 illustrates AI through:

Robotic vehicles

Speech recognition

Autonomous planning
and scheduling

Game playing

Spam filtering

Logistics planning

Robotics

Machine translation

These examples represent broader AI capabilities

Perception + Language + Planning + Learning + Action

Reference: T1, §1.4, pp. 28–29.

Integrated example: intelligent route planning

Problem: Recommend a route from home to the airport.

Stage	AI question
Perception	Where is the user? What are the current road conditions?
Representation	How should roads, intersections, distances and travel times be represented?
Search	Which possible routes should be explored?
Decision	Which route best satisfies the objective?
Action	What directions should be given?
Learning	How should predicted travel times improve using past journeys?

Home

Route A – ring road

Route B – city roads

selected

Route C – motorway

Airport

Possible objectives

- shortest distance
- least travel time
- lowest cost
- highest reliability

Instructor synthesis based on: T1, §§1.1 and 1.4; DSE ZG557 learning plan.



Activity: analyse an AI application

Choose one domain:

Healthcare

Finance

Manufacturing

Cybersecurity

Education

Transportation

Answer five questions

1. What information does the system receive?
2. What intelligent task does it perform?
3. What decision or action does it produce?
4. How should success be measured?
5. What can go wrong?

Reference: T1, §1.4, pp. 28–29; activity designed for LO1 and LO8.



What AI is and what it is not

AI is

- a scientific and engineering field
- concerned with perception, reasoning, learning and action
- based on explicit objectives and computational methods
- applicable to physical and digital systems

AI is not

- synonymous with ChatGPT
- synonymous with machine learning
- automatically intelligent merely because software is complex
- guaranteed to be correct, unbiased or explainable
- evidence that a machine thinks exactly like a human

AI is not magic: it is implemented through models, algorithms, data and computation.

Reference: T1, §§1.1 and 1.4, pp. 1–5 and 28–29.



Contact Session 1: key takeaways

1

AI can be approached in four ways

Thinking humanly | Acting humanly | Thinking rationally | Acting rationally

2

This course mainly adopts rational action

Choose the action expected to produce the best outcome given available information.

3

AI has interdisciplinary foundations

Philosophy, mathematics, economics, neuroscience, psychology, computer engineering, control theory and linguistics.

4

AI applications integrate multiple capabilities

Perception, representation, reasoning, decision, action and learning.

Reference: T1, Chapter 1 summary, pp. 29–30.



Exit check and preparation for CS2

Exit check

1. How is thinking rationally different from acting rationally?
2. Why does passing the Turing Test not prove that a machine thinks like a human?
3. Name three foundations of AI and one contribution from each.
4. Which AI approach best describes a system that chooses the lowest-cost route?

Preparation for Contact Session 2

Topic

Intelligent agents, environments and rationality

Required reading

T1, Chapter 2

Reference: DSE ZG557 Handout, Contact Session 2; T1, Chapter 2.

References



Primary course source

Birla Institute of Technology & Science, Pilani
Digital Learning Handout: Artificial and
Computational Intelligence
Course No. DSE ZG557, Version 5.0

Prescribed textbook

Stuart Russell and Peter Norvig
Artificial Intelligence: A Modern Approach
Third Edition

Sections used in Contact Session 1

§1.1 — What Is AI?
§1.2 — Foundations of Artificial Intelligence
§1.4 — State of the Art

Preparation for Contact Session 2

Chapter 2 — Intelligent Agents